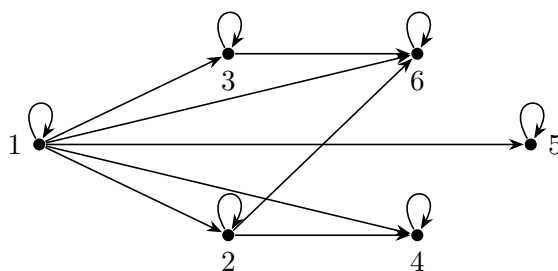


## 11 Relations

### 11.0 Relations

2. Let  $A = \{1, 2, 3, 4, 5, 6\}$ . Write out the relation  $R$  that expresses  $\mid$  (divides) on  $A$ . Then illustrate it with a diagram.

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$



6. Congruence modulo 5 is a relation on the set  $A = \mathbb{Z}$ . In this relation  $xRy$  means  $x \equiv y \pmod{5}$ . Write out the set  $R$  in set-builder notation.

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : 5 \mid (x - y)\}.$$

8. Let  $A = \{1, 2, 3, 4, 5, 6\}$ . Observe that  $\emptyset \subseteq A \times A$ , so  $R = \emptyset$  is a relation on  $A$ . Draw a diagram for this relation.

Since  $R = \emptyset$ , no pair  $(x, y)$  belongs to  $R$ , so the diagram has a node for each element of  $A$  and no arrows at all.



10. Consider the subset  $R = (\mathbb{R} \times \mathbb{R}) - \{(x, x) : x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$ . What familiar relation on  $\mathbb{R}$  is this? Explain.

The relation is *not equal* ( $\neq$ ). The set contains every pair  $(x, y)$  of real numbers. Removing the pair  $(x, x)$  leaves exactly the pairs where  $x$  and  $y$  differ.

## 11.1 Properties of Relations

2. Consider the relation  $R = \{(a, b), (a, c), (c, c), (b, b), (c, b), (b, c)\}$  on the set  $A = \{a, b, c\}$ . Is  $R$  reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is not reflexive because  $(a, a) \notin R$ .

This is not symmetric because  $(a, b) \in R$  but  $(b, a) \notin R$ .

This is transitive: checking every chain  $xRy, yRz$  of pairs in  $R$  confirms that  $(x, z) \in R$  in each case.

4. Let  $A = \{a, b, c, d\}$ . Suppose  $R$  is the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (a, c), (c, a), (a, d), (d, a), (b, c), (c, b), (b, d), (d, b), (c, d), (d, c)\}.$$

Is  $R$  reflexive? Symmetric? Transitive? If a property does not hold, say why.

This is reflexive because  $xRx$  holds for every  $x \in A$ .

This is symmetric because  $xRy$  and  $yRx$  both hold for every  $x, y \in A$ .

This is transitive because  $xRz$  holds for every  $x, z \in A$ , regardless of  $y$ .

6. Consider the relation  $R = \{(x, x) : x \in \mathbb{Z}\}$  on  $\mathbb{Z}$ . Is  $R$  reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Here  $xRy$  if and only if  $x = y$ , so  $R$  is the equality relation  $=$  on  $\mathbb{Z}$ .

This is reflexive because  $x = x$  for every  $x \in \mathbb{Z}$ .

This is symmetric because  $x = y$  implies  $y = x$ .

This is transitive because  $x = y$  and  $y = z$  imply  $x = z$ .

**8. Define a relation on  $\mathbb{Z}$  as  $xRy$  if  $|x - y| < 1$ . Is  $R$  reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?**

Since  $x - y$  is an integer and the only integer with absolute value less than 1 is 0, we have  $xRy$  if and only if  $x - y = 0$ , that is,  $x = y$ . So  $R$  is the equality relation  $=$  on  $\mathbb{Z}$ .

This is reflexive because  $x = x$  for every  $x \in \mathbb{Z}$ .

This is symmetric because  $x = y$  implies  $y = x$ .

This is transitive because  $x = y$  and  $y = z$  imply  $x = z$ .

**10. Suppose  $A \neq \emptyset$ . Since  $\emptyset \subseteq A \times A$ , the set  $R = \emptyset$  is a relation on  $A$ . Is  $R$  reflexive? Symmetric? Transitive? If a property does not hold, say why.**

This is not reflexive. Pick any  $x \in A$ , then  $(x, x) \notin R$  because  $R = \emptyset$ .

This is symmetric because it is impossible to find a  $(x, y) \in R$  for which  $(y, x) \notin R$ .

This is transitive because  $xRy \wedge yRz$  implies  $xRz$  for every  $x, y, z \in A$ .

**12. Prove that the relation  $|$  (divides) on the set  $\mathbb{Z}$  is reflexive and transitive. (Use Example 11.8 as a guide if you are unsure of how to proceed.)**

*Proof.* First we will show that  $|$  (divides) is reflexive. Take any integer  $x \in \mathbb{Z}$ . Since  $x = x \cdot 1$  and  $1 \in \mathbb{Z}$ , the definition of divisibility gives  $x | x$ . Thus  $|$  is reflexive.

Next, we will show that  $|$  is transitive. For this, we must show that if  $x | y$  and  $y | z$  then  $x | z$ . Suppose  $x | y$  and  $y | z$ . This means  $y = xm$  and  $z = yn$  for some integers  $m$  and  $n$ . Substituting  $xm$  for  $y$  we get  $z = (xm)n = x(mn)$ , and by definition of divisibility, we see  $x | z$ .

The past two paragraphs show that  $|$  is reflexive and transitive on  $\mathbb{Z}$ . □

**14. Suppose  $R$  is a symmetric and transitive relation on a set  $A$ , and there is an element  $a \in A$  for which  $aRx$  for every  $x \in A$ . Prove that  $R$  is reflexive.**

*Proof.* Let  $R$  be a transitive and symmetric relation in  $A$ . And assume there is an element  $a \in A$  such that  $aRx$  for every  $x \in A$ . We want to prove that  $xRx$  for every  $x \in A$ .

Take any  $x \in A$ . Since  $aRx$  for every  $x \in A$ , we have  $aRx$ . Because  $R$  is symmetric, we have  $xRa$ . So we have both  $xRa$  and  $aRx$ . Because  $R$  is transitive,  $xRa$  and  $aRx$  imply  $xRx$ . Therefore,  $R$  is transitive, symmetric, and reflexive. □

**16. Define a relation  $R$  on  $\mathbb{Z}$  by declaring that  $xRy$  if and only if  $x^2 \equiv y^2 \pmod{4}$ . Prove that  $R$  is reflexive, symmetric and transitive.**

*Proof.* Assume  $R$  is a relation on  $\mathbb{Z}$  such as  $xRy$  if and only if  $x^2 \equiv y^2 \pmod{4}$ . We want to prove that  $R$  is reflexive, symmetric, and transitive.

First, we will show that  $R$  is reflexive. Take any integer  $x \in \mathbb{Z}$ , and observe that  $4 \mid 0$ , so  $4 \mid x^2 - x^2$ . Thus, by definition,  $x^2 \equiv x^2 \pmod{4}$ . This shows  $R$  is reflexive.

Second, we will show that  $R$  is symmetric. Suppose  $xRy$  for some  $x, y \in \mathbb{Z}$ . So  $4 \mid (x^2 - y^2)$ . This means  $(x^2 - y^2) = 4m$  for some  $m \in \mathbb{Z}$ . Then  $y^2 - x^2 = 4(-m)$ , so  $4 \mid (y^2 - x^2)$ . Thus by definition,  $y^2 \equiv x^2 \pmod{4}$ . Therefore  $R$  is symmetric.

Finally, we will show that  $R$  is transitive. Suppose  $xRy$  and  $yRz$  for  $x, y, z \in \mathbb{Z}$ . So  $x^2 - y^2 = 4k$  for some integer  $k$  and  $y^2 - z^2 = 4l$  for some integer  $l$ . Adding these two equations, we obtain  $x^2 - z^2 = 4(k + l)$ , so  $4 \mid (x^2 - z^2)$ . Hence  $x^2 \equiv z^2 \pmod{4}$ . Therefore  $R$  is transitive.

The past three paragraphs have shown that  $R$  is reflexive, symmetric and transitive.  $\square$

## 11.2 Equivalence Relations

**2. Let  $A = \{a, b, c, d, e\}$ . Suppose  $R$  is an equivalence relation on  $A$ . Suppose  $R$  has two equivalence classes. Also  $aRd$ ,  $bRc$  and  $eRd$ . Write out  $R$  as a set.**

Since  $aRd$  and  $eRd$ , the elements  $a$ ,  $d$  and  $e$  lie in one equivalence class, and since  $bRc$ , the elements  $b$  and  $c$  lie in another. These are the two classes:  $[a] = [d] = [e] = \{a, d, e\}$  and  $[b] = [c] = \{b, c\}$ . Each class  $X$  contributes the pairs  $X \times X$  to  $R$ , so

$$R = \{(a, a), (d, d), (e, e), (a, d), (d, a), (a, e), (e, a), (d, e), (e, d), (b, b), (c, c), (b, c), (c, b)\}.$$

**4. Let  $A = \{a, b, c, d, e\}$ . Suppose  $R$  is an equivalence relation on  $A$ . Suppose also that  $aRd$  and  $bRc$ ,  $eRa$  and  $cRe$ . How many equivalence classes does  $R$  have?**

$R$  has one equivalence class. Since  $aRd$  and  $eRa$ , the elements  $a$ ,  $d$  and  $e$  lie in one class, and since  $bRc$  and  $cRe$ , the elements  $b$ ,  $c$  and  $e$  lie in one class. As  $e$  belongs to both, all five elements share a single class:  $[a] = [b] = [c] = [d] = [e] = A$ .

**6. There are five different equivalence relations on the set  $A = \{a, b, c\}$ . Describe them all. Diagrams will suffice.**

$$\begin{aligned} R_1 &= \{(a, a), (b, b), (c, c)\} \\ R_2 &= \{(a, a), (b, b), (c, c), (a, b), (b, a)\} \end{aligned}$$

$$\begin{aligned}
R_3 &= \{(a, a), (b, b), (c, c), (a, c), (c, a)\} \\
R_4 &= \{(a, a), (b, b), (c, c), (b, c), (c, b)\} \\
R_5 &= \{(a, a), (b, b), (c, c), (a, b), (b, a), (a, c), (c, a), (b, c), (c, b)\}.
\end{aligned}$$

8. Define a relation  $R$  on  $\mathbb{Z}$  as  $xRy$  if and only if  $x^2 + y^2$  is even. Prove  $R$  is an equivalence relation. Describe its equivalence classes.

10. Suppose  $R$  and  $S$  are two equivalence relations on a set  $A$ . Prove that  $R \cap S$  is also an equivalence relation. (For an example of this, look at Figure 11.2. Observe that for the equivalence relations  $R_2, R_3$  and  $R_4$ , we have  $R_2 \cap R_3 = R_4$ .)

12. Prove or disprove: If  $R$  and  $S$  are two equivalence relations on a set  $A$ , then  $R \cup S$  is also an equivalence relation on  $A$ .

14. Suppose  $R$  is a reflexive and symmetric relation on a finite set  $A$ . Define a relation  $S$  on  $A$  by declaring  $xSy$  if and only if for some  $n \in \mathbb{N}$  there are elements  $x_1, x_2, \dots, x_n \in A$  satisfying  $xRx_1, x_1Rx_2, x_2Rx_3, x_3Rx_4, \dots, x_{n-1}Rx_n$ , and  $x_nRy$ . Show that  $S$  is an equivalence relation and  $R \subseteq S$ . Prove that  $S$  is the unique smallest equivalence relation on  $A$  containing  $R$ .

### 11.3 Equivalence Classes and Partitions

2. List all the partitions of the set  $A = \{a, b, c\}$ . Compare your answer to the answer to Exercise 6 of Section 11.2.

4. Suppose  $P$  is a partition of a set  $A$ . Define a relation  $R$  on  $A$  by declaring  $xRy$  if and only if  $x, y \in X$  for some  $X \in P$ . Prove  $R$  is an equivalence relation on  $A$ . Then prove that  $P$  is the set of equivalence classes of  $R$ .

### 11.4 The Integers Modulo $n$

2. Write the addition and multiplication tables for  $\mathbb{Z}_3$ .

4. Write the addition and multiplication tables for  $\mathbb{Z}_6$ .

6. Suppose  $[a], [b] \in \mathbb{Z}_6$  and  $[a] \cdot [b] = [0]$ . Is it necessarily true that either  $[a] = [0]$  or  $[b] = [0]$ ?

8. Suppose  $[a], [b] \in \mathbb{Z}_n$ , and  $[a] = [a']$  and  $[b] = [b']$ . Alice adds  $[a]$  and  $[b]$  as  $[a] + [b] = [a + b]$ . Bob adds them as  $[a'] + [b'] = [a' + b']$ . Show that their answers  $[a + b]$  and  $[a' + b']$  are the same.